■ beacon

v0.1 · DRAFT

Open-Source AI Research Agent for GitHub Repositories

Author Product Team	Status Draft	Date June 2026	Audience Engineering · OSS Contributors

01 · OVERVIEW

Problem & Vision

Understanding an unfamiliar GitHub repository today means hours of manual work — reading docs, browsing issues, inspecting PRs, and guessing which maintainer to contact. **Beacon** automates this research end-to-end. It collects repository metadata, source structure, issues, pull requests, review history, contributor activity, and discussions, then uses AI reasoning to produce **structured, contributor-tailored intelligence** — not generic summaries.

Long-term, Beacon becomes the open-source research layer for software development: reducing onboarding friction, accelerating contributions, and making open source more accessible at every skill level.

02 · TARGET USERS

Who Beacon Serves

User Segment	Primary Need	Key Value
New Contributors	Identify realistic entry-point issues	Genuine beginner-friendly issue ranking beyond labels
Experienced Developers	Deep codebase intelligence before starting work	Architecture maps, maintainer routing, PR patterns
Maintainers / OSS Leads	Monitor project health & contributor activity	Contribution analytics, review bottleneck detection

03 · CORE FEATURES

What Beacon Does

Issue Intelligence	Ranks issues by real approachability — analysing complexity, code ownership, review history, and required knowledge — not just `good-first-issue` labels.
Repository Architecture Map	Explains module structure, component relationships, key subsystems, and maps which maintainers own which code paths.
PR Pattern Matching	Finds prior merged PRs similar to the current task, surfaces implementation patterns used and reviews received.

Natural-Language Q&A	Answers developer questions grounded in indexed repo data: which files, which maintainer, which tests, what to read first.
Contributor Intelligence	Generates personalised issue recommendations, learning paths, suggested reading order, and implementation guidance.
Project Health Dashboard	Summarises release cadence, review latency, bus-factor risk, and overall contributor activity trends.

04 · SYSTEM ARCHITECTURE

Modular Components

Component	Responsibility	Primary APIs / Tech
Repository Collector	Fetches metadata, source tree, issues, PRs, commits, contributors, discussions	GitHub REST + GraphQL APIs
Knowledge Index	Structures and indexes collected data for semantic retrieval	Vector DB (Chroma / pgvector), embeddings
Research Engine	Orchestrates specialised agents: architecture, issue, PR, maintainer, health	LLM agents, tool-calling
Interactive Assistant	Answers natural-language questions grounded in indexed repo data	RAG pipeline, LLM
Contributor Intelligence	Generates personalised recommendations, issue rankings, learning paths	Scoring models, LLM

05 · EXAMPLE QUERIES

Questions Beacon Answers

- Which files are involved in this issue?
- Which maintainer usually reviews authentication-related code?
- Has this problem been solved before? What approach was used?
- Which tests should I update when changing this module?
- What should I read before working on this feature?
- Is this a healthy project to contribute to? What's the review turnaround?

06 · ROADMAP

Phased Delivery

Phase	Milestone	Scope	Priority
α · v0.1	Core Data Pipeline	Collector + Knowledge Index + CLI output	● High
β · v0.2	Research Engine	Architecture agent, issue ranker, PR pattern matcher	● High
v0.3	Interactive Assistant	RAG Q&A;, natural-language interface	■ Med
v0.4	Contributor Intelligence	Personalised recommendations, learning paths	■ Med
v1.0	Hosted Platform	Managed infra, continuous monitoring, enterprise integrations	■ Low

07 · SCOPE & NON-GOALS

What Beacon Is Not

- A code generation or auto-PR tool — Beacon researches, it does not write code.
- A replacement for GitHub UI — it augments, not replaces, the standard workflow.
- Limited to GitHub — the modular design supports other platforms in future (GitLab, Bitbucket).
- A one-size report — output is tailored by contributor skill level and stated goal.

08 · SUCCESS METRICS

How We Measure Impact

Metric	Definition
Time-to-first-contribution	Reduction in days from repo discovery to opened PR
Issue recommendation accuracy	% of recommended issues successfully resolved by target user tier
Q&A; grounding quality	Factual accuracy score vs manual ground truth (LLM-as-judge)
OSS adoption	GitHub stars, forks, and active installations at 6 months